```python
import pandas as pd

df = pd.read_csv('/content/data_science.csv')

print('First 5 rows of the dataset:')
display(df.head()[:7])

print('\nDataset Info:')
df.info()
```

```
/tmp/ipykernel_2873/3097406522.py:3: DtypeWarning: Columns (9) have mixed types. Specify dtype option on import or set low_memory=False.
  df = pd.read_csv('/content/data_science.csv')
First 5 rows of the dataset:
```

	id	conversation_id	created_at	date	time	timezone	user_id	username	name	place	...	geo	source	user_rt_id	user_rt	retweet_id	reply_to	retweet_date	translate
0	1406400408545804288	1406400396264943616	2021-06-20 05:26:01 IST	2021-06-20	05:26:01	530	1113747629282930688	ballouxfrancois	Prof Francois Balloux	NaN	...	NaN	NaN	NaN	NaN	NaN	[]	NaN	NaN
1	1406390341176016897	1406390341176016897	2021-06-20 04:46:01 IST	2021-06-20	04:46:01	530	788898706586275840	tdatascience	Towards Data Science	NaN	...	NaN	NaN	NaN	NaN	NaN	[]	NaN	NaN
2	1406386311481774083	1406386311481774083	2021-06-20 04:30:00 IST	2021-06-20	04:30:00	530	19402238	sciencenews	Science News	NaN	...	NaN	NaN	NaN	NaN	NaN	[]	NaN	NaN
3	1406383545153638402	1406383545153638402	2021-06-20 04:19:01 IST	2021-06-20	04:19:01	530	788898706586275840	tdatascience	Towards Data Science	NaN	...	NaN	NaN	NaN	NaN	NaN	[]	NaN	NaN
4	1406358632648818689	1406358632648818689	2021-06-20 02:40:01 IST	2021-06-20	02:40:01	530	788898706586275840	tdatascience	Towards Data Science	NaN	...	NaN	NaN	NaN	NaN	NaN	[]	NaN	NaN

5 rows × 36 columns

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 241386 entries, 0 to 241385
Data columns (total 36 columns):
 #   Column           Non-Null Count   Dtype
---  ------           --------------   -----
 0   id               241386 non-null  int64
 1   conversation_id  241386 non-null  int64
 2   created_at       241386 non-null  object
 3   date             241386 non-null  object
 4   time             241386 non-null  object
 5   timezone         241386 non-null  int64
 6   user_id          241386 non-null  int64
 7   username         241386 non-null  object
 8   name             241386 non-null  object
 9   place            354 non-null     object
 10  tweet            241386 non-null  object
 11  language         241386 non-null  object
 12  mentions         241386 non-null  object
 13  urls             241386 non-null  object
 14  photos           241386 non-null  object
 15  replies_count    241386 non-null  int64
 16  retweets_count   241386 non-null  int64
 17  likes_count      241386 non-null  int64
 18  hashtags         241386 non-null  object
 19  cashtags         241386 non-null  object
 20  link             241386 non-null  object
 21  retweet          241386 non-null  bool
 22  quote_url        10321 non-null   object
 23  video            241386 non-null  int64
 24  thumbnail        110338 non-null  object
 25  near             0 non-null       float64
 26  geo              0 non-null       float64
 27  source           0 non-null       float64
 28  user_rt_id       0 non-null       float64
 29  user_rt          0 non-null       float64
 30  retweet_id       0 non-null       float64
 31  reply_to         241386 non-null  object
 32  retweet_date     0 non-null       float64
 33  translate        0 non-null       float64
 34  trans_src        0 non-null       float64
 35  trans_dest       0 non-null       float64
dtypes: bool(1), float64(10), int64(8), object(17)
memory usage: 64.7+ MB
```

```python
import re
import string

def clean_text(text):
    text = text.lower()
    text = re.sub(r'https?://\S+|www\.\S+', '', text) # Remove URLs
    text = re.sub(r'<.*?>+', '', text) # Remove HTML tags
    text = re.sub(r'[%s]' % re.escape(string.punctuation), '', text) # Remove punctuation
    text = re.sub(r'\n', '', text) # Remove newlines
    text = re.sub(r'\w*\d\w*', '', text) # Remove words containing numbers
    text = re.sub(r'@\w+', '', text) # Remove mentions
    text = re.sub(r'#\w+', '', text) # Remove hashtags
    return text

df['cleaned_tweet'] = df['tweet'].apply(clean_text)

print('Original Tweet vs. Cleaned Tweet:')
display(df[['tweet', 'cleaned_tweet']].head())
```

```
Original Tweet vs. Cleaned Tweet:
```

	tweet	cleaned_tweet
0	What can be done? - Never blindly trust an ab...	what can be done never blindly trust an abst...
1	"We need a paradigm shift from model-centric t...	we need a paradigm shift from modelcentric to ...
2	Using high-resolution satellite data and compu...	using highresolution satellite data and comput...
3	.@Stephenson_Data shares four steps that will ...	stephensondata shares four steps that will hel...
4	"Curricula is inherently brittle in a world wh...	curricula is inherently brittle in a world whe...

```
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

nltk.download('punkt')
nltk.download('stopwords')
nltk.download('punkt_tab')

stop_words = set(stopwords.words('english'))

def tokenize_and_remove_stopwords(text):
    tokens = word_tokenize(text)
    filtered_tokens = [word for word in tokens if word not in stop_words]
    return ' '.join(filtered_tokens)

df['processed_tweet'] = df['cleaned_tweet'].apply(tokenize_and_remove_stopwords)

print('\nCleaned Tweet vs. Processed Tweet (Tokenized and Stopwords Removed):')
display(df[['cleaned_tweet', 'processed_tweet']].head())
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.

Cleaned Tweet vs. Processed Tweet (Tokenized and Stopwords Removed):
```

	cleaned_tweet	processed_tweet
0	what can be done never blindly trust an abst...	done never blindly trust abstract press releas...
1	we need a paradigm shift from modelcentric to ...	need paradigm shift modelcentric datacentric m...
2	using highresolution satellite data and comput...	using highresolution satellite data computer a...
3	stephensondata shares four steps that will hel...	stephensondata shares four steps help new data...
4	curricula is inherently brittle in a world whe...	curricula inherently brittle world indemand sk...

```
from sklearn.feature_extraction.text import TfidfVectorizer

tfidf_vectorizer = TfidfVectorizer(max_features=5000)
tfidf_matrix = tfidf_vectorizer.fit_transform(df['processed_tweet'])

print('TF-IDF matrix shape:', tfidf_matrix.shape)
```

```
TF-IDF matrix shape: (241386, 5000)
```

```
import nltk
nltk.download('vader_lexicon')
from nltk.sentiment.vader import SentimentIntensityAnalyzer

sia = SentimentIntensityAnalyzer()

def get_vader_sentiment(text):
    score = sia.polarity_scores(text)
    if score['compound'] >= 0.05:
        return 'positive'
    elif score['compound'] <= -0.05:
        return 'negative'
    else:
        return 'neutral'

df['sentiment'] = df['processed_tweet'].apply(get_vader_sentiment)

print('\nSentiment distribution:')
display(df['sentiment'].value_counts())
display(df[['processed_tweet', 'sentiment']].head())
display(df[['processed_tweet', 'sentiment']].tail())
```

```
[nltk_data] Downloading package vader_lexicon to /root/nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!

Sentiment distribution:
```

sentiment	count
positive	121602
neutral	97629
negative	22155

dtype: int64

	processed_tweet	sentiment
0	done never blindly trust abstract press releas...	negative
1	need paradigm shift modelcentric datacentric m...	neutral
2	using highresolution satellite data computer a...	neutral
3	stephensondata shares four steps help new data...	positive
4	curricula inherently brittle world indemand sk...	positive

	processed_tweet	sentiment
241381	cda jobs data dec employment rose health socia...	neutral
241382	rt filiber computer science background love de...	positive
241383	heck science ive got empirical data heh heh	negative
241384	thedata rt noahwg dr petra provides perspectiv...	negative
241385	“ world retail always mix art science ” nyt da...	neutral